EXP 20

PERFORMANCE EVALUATION OF FREE-SPACE OPTICAL (FSO) LINK UNDER ATMOSPHERIC TURBULENCE USING OPTISYSTEM

OBJECTIVES

1. Analyze the effect of range on received power, considering the combined contributions of geometric spreading loss and atmospheric attenuation

2. Evaluate how the combined effects of increasing range and attenuation influence received power and bit error rate (BER).

3. Assess the extent to which increased transmit power can compensate for attenuation-induced signal loss.

EXPECTED OUTCOME

Students shall be able to:

1. Received power decreases with range due to geometric spreading and atmospheric a attenuation, with attenuation dominating at longer distances.

2. Increasing range and attenuation degrade received power and sharply increase BER beyond a certain range, defining a maximum reliable communication distance.

3. Higher transmit power can partially compensate for attenuation and extend range, but with diminishing returns at high attenuation levels.

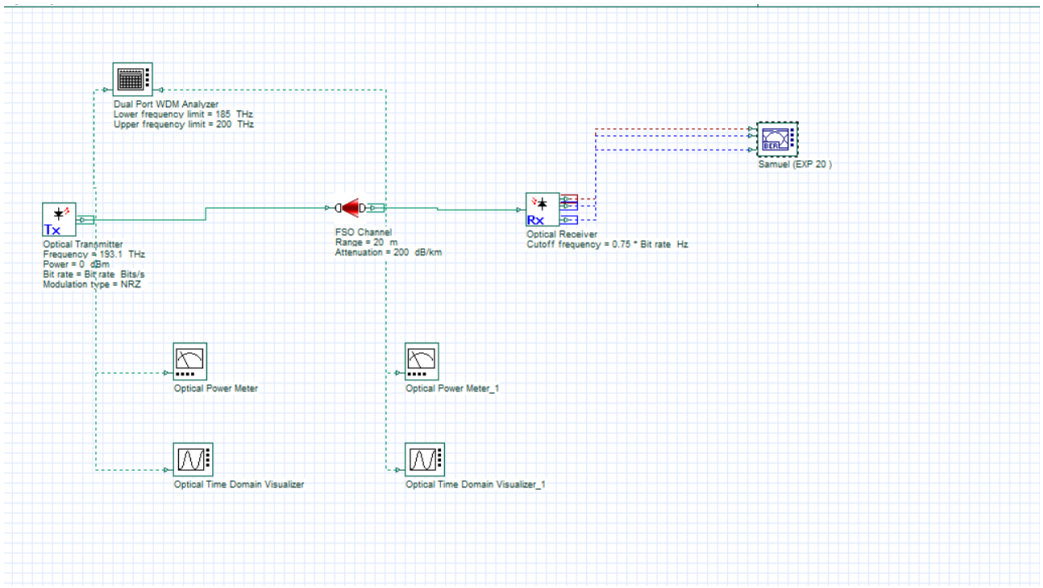
SOFTWARE REQUIRED

• Optisystem

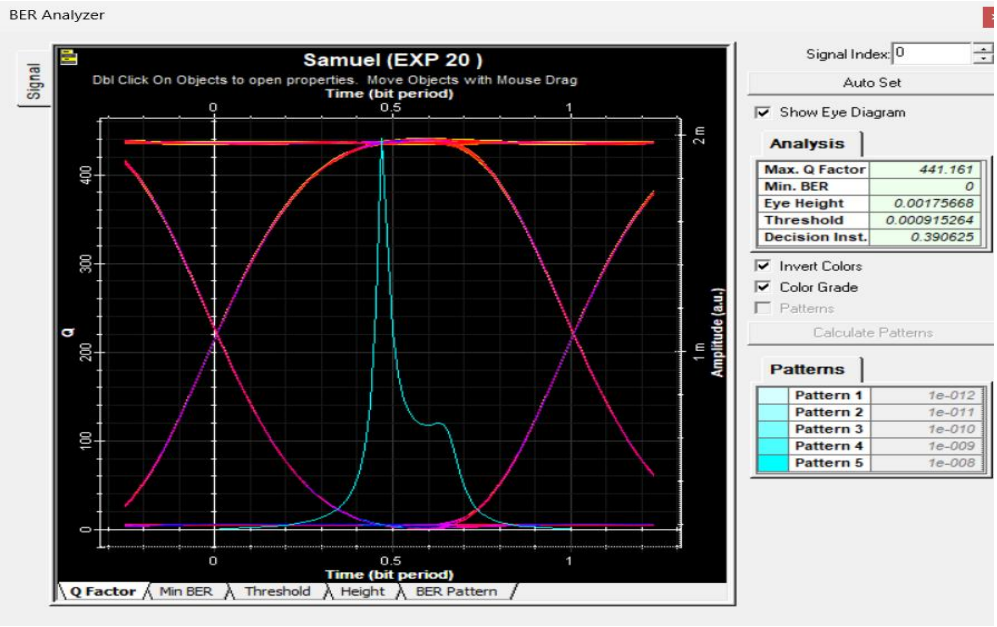
EXPERIMENT PROCEDURE

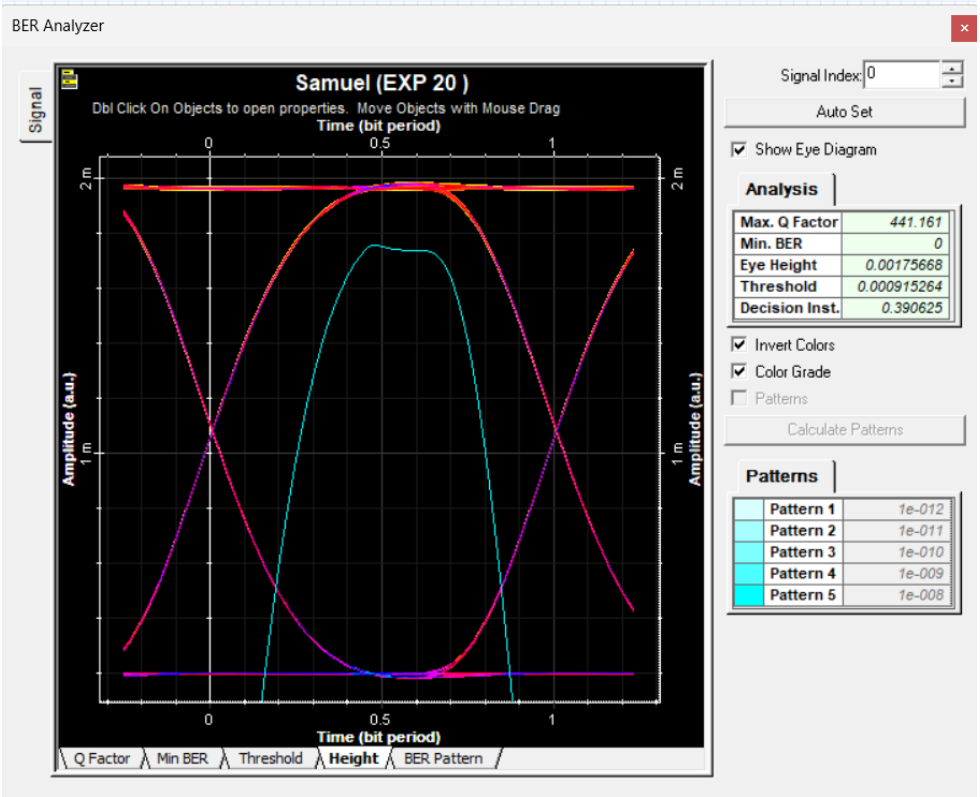
1. Design the FSO transmi er and receiver in OptiSystem using suitable optical source, modulator, photodetector, and receiver components.
2. Set input parameters: bit rate, data pa ern, optical power, and wavelength.
3. Connect transmi er, FSO channel, and receiver; configure transmission range and a enuation values.
4. Connect BER Analyzer, Eye Diagram Analyzer, and Optical Power Meter to monitor performance.
5. Vary range and a enuation, and simulate each condition.
6. Record BER, Q-factor, eye diagram, and received power for each case.
7. Compare results to analyze the effect of range and a enuation on FSO link performance.

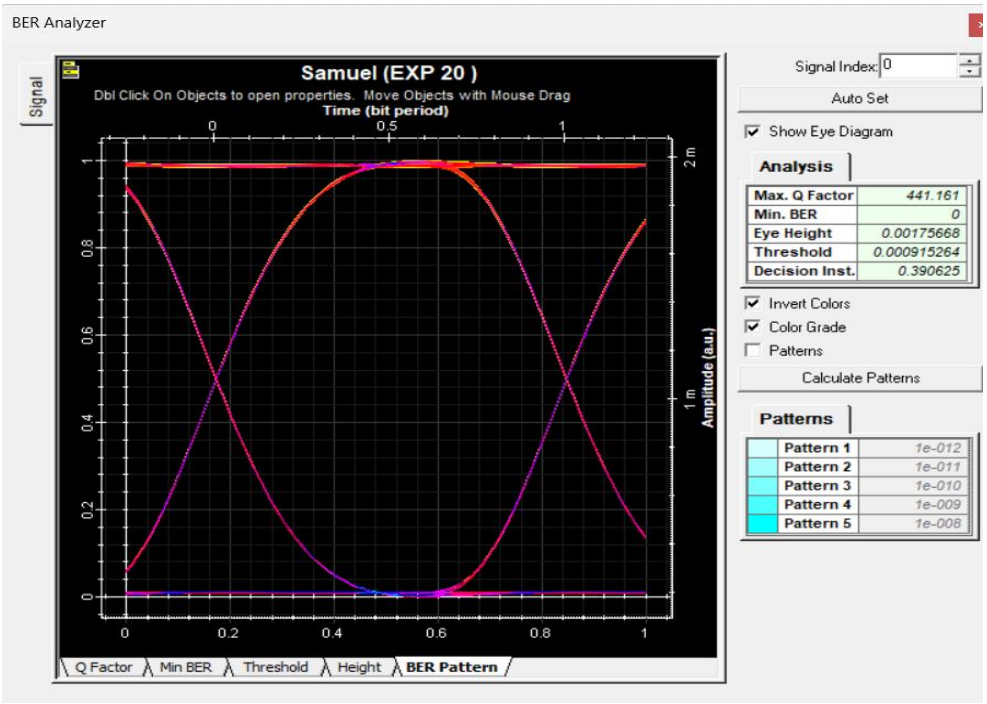
BLOCK DIAGRAM



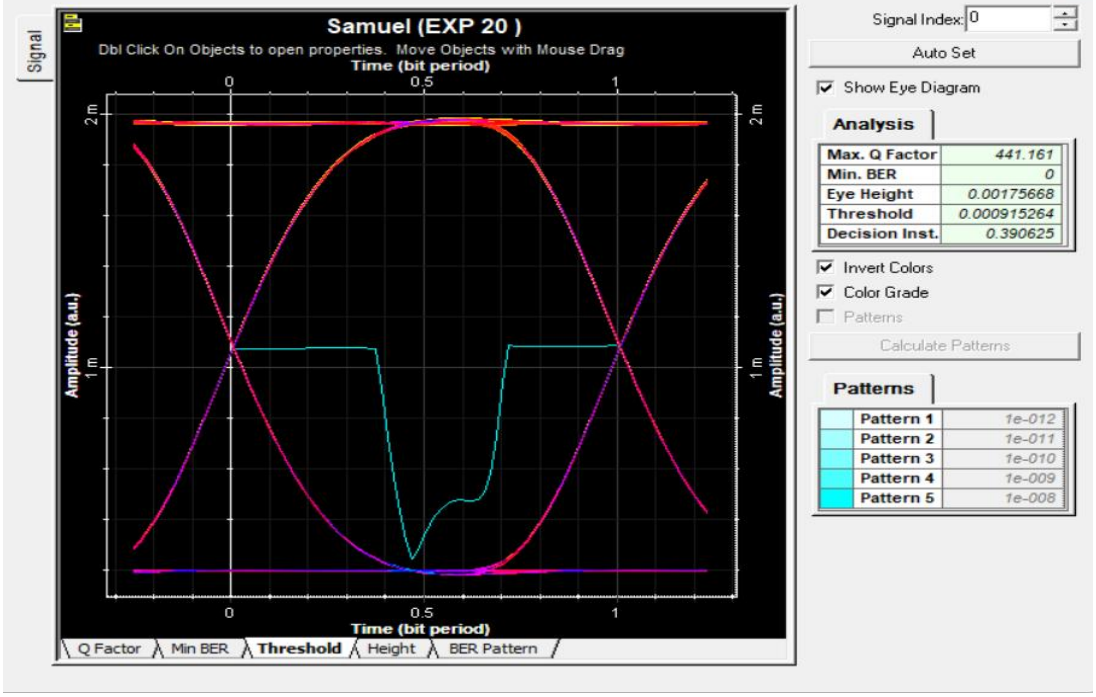
OUTPUT :



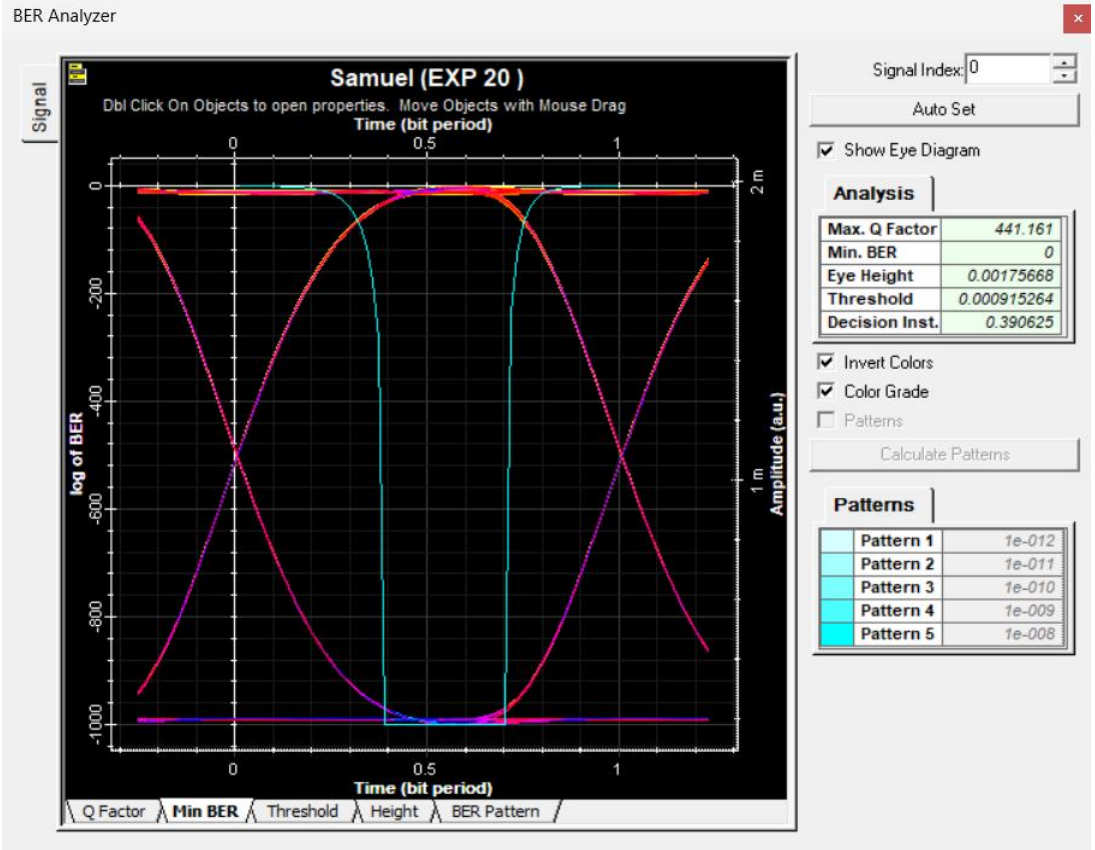

BER Analyzer
Signal
Samuel (EXP 20)
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Amplitude (a.u.)
Time (bit period)
Q Factor
Min BER
Threshold
Height
BER Pattern
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 441.161
Min. BER 0
Eye Height 0.00175668
Threshold 0.000915264
Decision Inst. 0.390625
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008


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BER Analyzer



BER Analyzer



LABORATORY PERFORMANCE ASSESSMENT RUBRIC

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Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & FSO Link Setup	Correct selection and connection of OptiSystem components for Free Space Optics link (Optical Transmitter, FSO Channel, PIN Photodetector, Low Pass Filter, Optical Power Meter, Time Domain Visualizer, BER Analyzer).	All optical transmitter, free space channel, receiver, and visualizer components are correctly selected, connected in sequence, and configured without errors.	All components are present with minor connection or layout errors, but the FSO link remains functional.	Component selection is partially complete; minor connection or configuration errors impede optical signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of physical input parameters: Wavelength (1550 nm), Modulation (OOK/NRZ), Transmit Power, FSO Range (1 to 15 km), and Atmospheric Attenuation (2 to 10 dB/km).	All input parameters and test cases across varying ranges and attenuation values are accurately set and executed.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but multiple range or attenuation test cases are missing.	Incorrect input parameter settings or failure to run the required atmospheric test cases.	20%
Measurement & Power Monitoring	Accurate extraction and recording of received optical power, Q-Factor, BER, Eye Height, and Threshold across all range settings.	Precise data extraction across all FSO range cases, neatly tabulated with complete optical power readings and clear time-domain waveforms.	Data is collected with minor omissions in recording power meter readings, BER, or eye parameters.	Partial data collection with incomplete tables or missing optical power visualizer readings.	Inaccurate, incomplete, or missing data and measurements.	20%
Atmospheric & Link Loss Analysis	Detailed evaluation of geometric spreading loss, atmospheric attenuation, distance-dependent signal degradation, and power compensation feasibility.	Comprehensive analysis correctly relating range and attenuation variations to geometric loss, received power drop, eye height reduction, and bit error rate performance.	Clear analysis identifying basic trends in received power and eye height relative to FSO distance.	Basic attempt at result analysis without explaining physical mechanisms like atmospheric scattering or geometric loss.	Erroneous or missing analysis of simulation results and power meter outputs.	20%
Report & Conclusion	Identification of maximum reliable communication distance, power compensation trade-offs with diminishing returns, and structural clarity.	Report features clear time-domain waveforms, complete power tables, and insightful conclusions identifying optimal operational parameters under atmospheric impairments.	Satisfactory report with relevant visualizer displays and sound concluding summary.	Incomplete report with missing graphs, power readings, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%